Math 110 Midterm 2 Drills

Solutions

"Prop. n" refers to the lecture notes. E_{ij} is the matrix with a 1 in entry (i,j) and zeros elsewhere.

§1 Products, affine subsets and quotient spaces

1. A product space.

a) $(1,0),\ (x,0),\ (0,e_1),\ (0,e_2)$, where 0 denotes the zero of the other factor; $\dim V = 2+2 = 4$.

b) T is linear (each entry enters linearly). Injective: if $p+ax^2+bx^3 = 0$ with $\deg p \le 1$, comparing coefficients gives $a = b = 0$ and $p = 0$. Since $\dim V = 4 = \dim \mathcal{P}_3(\mathbb{R})$, an injective linear map between them is an isomorphism.

c) It is the null space of the functional $\varphi\big(p,(a,b)\big) = p(1) - a - b$, which is nonzero ($\varphi(1,(0,0)) = 1$). A nonzero functional has null space of dimension $\dim V - 1 = 3$.

2. The sum map.

a) $\Gamma\big((u_i)+(u_i')\big) = \sum(u_i+u_i') = \sum u_i + \sum u_i'$ and $\Gamma(\lambda(u_i)) = \lambda\sum u_i$.

b) Every value $u_1+\cdots+u_m$ lies in $U_1+\cdots+U_m$, and every element of $U_1+\cdots+U_m$ has this form.

c) Γ is injective iff $\operatorname{null}\Gamma = \{0\}$, i.e. iff $u_1+\cdots+u_m = 0$ with $u_i \in U_i$ forces every $u_i = 0$. That is exactly the condition for the sum to be direct.

d) Rank–nullity for Γ: $\dim(U_1+\cdots+U_m) = \dim(U_1\times\cdots\times U_m) - \dim\operatorname{null}\Gamma = \sum\dim U_i - \dim\operatorname{null}\Gamma$. So $\le$ holds, with equality iff $\operatorname{null}\Gamma = \{0\}$, iff the sum is direct by (c).

3. Affine subsets.

a) For $u \in U$: $v+u = w+\big((v-w)+u\big) \in w+U$. So $v+U \subseteq w+U$. Since $w-v = -(v-w) \in U$, the same argument gives $w+U \subseteq v+U$.

b) Pick $v+u_1 = w+u_2$ in the intersection. Then $v-w = u_2-u_1 \in U$.

c) $v \in x+W$, so $v = x+w_0$ with $w_0 \in W$. For $u \in U$, $v+u \in x+W$, so $v+u = x+w$ for some $w \in W$, and $u = w-w_0 \in W$. So $U \subseteq W$; swap the roles to get $W \subseteq U$.

d) $U = \{(x,y,z) : x+2y-z = 0\}$ and $v = (3,0,0)$: $(x,y,z) \in A$ iff $(x,y,z)-(3,0,0) \in U$.

4. Solution sets.

a) $x+y+z = 0$, $x-z = 0$: $x = z$, $y = -2z$. $\operatorname{null} T = \operatorname{span}\big((1,-2,1)\big)$.

b) $(1,1,0)$ works: $1+1+0 = 2$ and $1-0 = 1$. So the set is $(1,1,0)+\operatorname{span}\big((1,-2,1)\big)$.

c) If $Tu_0 = b$, then $Tu = b$ iff $T(u-u_0) = 0$ iff $u-u_0 \in \operatorname{null} T$ iff $u \in u_0+\operatorname{null} T$.

5. Operations on V/U.

a) $v-v' \in U$ and $w-w' \in U$, so $(v+w)-(v'+w') = (v-v')+(w-w') \in U$, so $(v+w)+U = (v'+w')+U$ (Q3(a)).

b) $\lambda v - \lambda v' = \lambda(v-v') \in U$.

c) $0+U \in W/U$ since $0 \in W$; $(w_1+U)+(w_2+U) = (w_1+w_2)+U$ and $\lambda(w+U) = \lambda w+U$ stay in W/U since W is a subspace. W/U is the quotient of the space W by its subspace U, so $\dim W/U = \dim W - \dim U$ (Prop. 40 applied to W).

d) $(1,2,5)-(1,2,-1) = (0,0,6) \in U$: equal. $(1,2,5)-(1,3,5) = (0,-1,0) \notin U$: not equal.

6. A basis of a quotient.

a) $(1,1,0,0), (0,0,1,1), e_1, e_3$ is a basis of $\mathbb{R}^4$ (it spans: $e_2 = (1,1,0,0) - e_1$, $e_4 = (0,0,1,1) - e_3$). So $e_1 + U$, $e_3 + U$ is a basis of V/U.

b) $(3,1,4,1) = 1 \cdot (1,1,0,0) + 1 \cdot (0,0,1,1) + 2e_1 + 3e_3$, so $(3,1,4,1) + U = 2(e_1 + U) + 3(e_3 + U)$.

c) $T(x_1, x_2, x_3, x_4) = (x_1 - x_2,\ x_3 - x_4)$: $\operatorname{null} T = \{x_1 = x_2,\ x_3 = x_4\} = U$, and T is surjective ($Te_1 = (1,0)$, $Te_3 = (0,1)$). Isomorphism: $x + U \mapsto (x_1 - x_2, x_3 - x_4)$.

7. Quotients of an infinite-dimensional space.

a) $p \mapsto p(0)$ is linear, surjective onto $\mathbb{R}$, with null space U. By the first isomorphism theorem $V/U \cong \mathbb{R}$, so $\dim V/U = 1$; basis $1 + U$ (it maps to 1).

b) $p \mapsto \big(p(0), p'(0)\big)$ is surjective onto $\mathbb{R}^2$ ($1 \mapsto (1,0)$, $x \mapsto (0,1)$) with null space W. So $V/W \cong \mathbb{R}^2$ and $\dim V/W = 2$.

c) $\pi\colon V \to V/\{0\}$ is surjective with null space $\{0\}$, so it is an isomorphism ($p \mapsto p + \{0\}$).

Note that V is infinite-dimensional but these quotients are finite-dimensional.

§2 Quotient maps and the isomorphism theorems

8. The first isomorphism theorem.

a) If $v + \operatorname{null} T = v' + \operatorname{null} T$, then $v - v' \in \operatorname{null} T$, so $Tv - Tv' = T(v - v') = 0$.

b) $\bar{T}\big((v + \operatorname{null} T) + \lambda(w + \operatorname{null} T)\big) = \bar{T}\big((v + \lambda w) + \operatorname{null} T\big) = T(v + \lambda w) = Tv + \lambda Tw$.

c) If $\bar{T}(v + \operatorname{null} T) = 0$, then $Tv = 0$, so $v \in \operatorname{null} T$ and $v + \operatorname{null} T = 0 + \operatorname{null} T$.

d) Each $Tv \in \operatorname{range} T$ equals $\bar{T}(v + \operatorname{null} T)$. So $\bar{T}\colon V/\operatorname{null} T \to \operatorname{range} T$ is an isomorphism. Differentiation D on $\mathcal{P}(\mathbb{R})$ is surjective (every polynomial has an antiderivative) with $\operatorname{null} D = \mathbb{R}$, so $\mathcal{P}(\mathbb{R})/\mathbb{R} \cong \mathcal{P}(\mathbb{R})$ via $p + \mathbb{R} \mapsto p'$.

9. Symmetric parts.

a) $\operatorname{null} T = \{A : A^T = -A\} = \operatorname{span}\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. $\operatorname{range} T$ is the set of symmetric matrices: $(A + A^T)^T = A + A^T$, and a symmetric S equals $T(\frac{1}{2}S)$.

b) $A + \operatorname{null} T \mapsto A + A^T$.

c) $E_{11} + \operatorname{null} T$, $E_{22} + \operatorname{null} T$, $E_{12} + \operatorname{null} T$, with images $2E_{11}$, $2E_{22}$, $E_{12} + E_{21}$ (a basis of the symmetric matrices, so the three cosets form a basis of $V/\operatorname{null} T$).

d) $\dim V/\operatorname{null} T = 4 - 1 = 3 = \dim \operatorname{range} T$.

10. Taylor coefficients.

a) $T(1) = (1,0,0)$, $T(x-2) = (0,1,0)$, $T\big((x-2)^2\big) = (0,0,2)$, so T is surjective and $\dim U = 5 - 3 = 2$. $(x-2)^3$ and $(x-2)^3 x$ lie in U (each has $p(2) = p'(2) = p''(2) = 0$) and are independent, so they span U: $U = \{(x-2)^3 q : q \in \mathcal{P}_1(\mathbb{R})\}$.

b) $p + U \mapsto \big(p(2), p'(2), p''(2)\big)$.

c) Their images $(1,0,0), (0,1,0), (0,0,2)$ form a basis of $\mathbb{R}^3$, and an isomorphism carries bases to bases.

d) $x^4 \mapsto (16, 32, 48) = 16(1,0,0) + 32(0,1,0) + 24(0,0,2)$. So $x^4 + U = 16(1 + U) + 32\big((x-2) + U\big) + 24\big((x-2)^2 + U\big)$.

11. Maps out of a quotient.

a) Well defined: $v - v' \in U \subseteq \operatorname{null} T$ gives $Tv = Tv'$. Linear: from T. Unique: any such map has the prescribed value at every element $v + U$ of V/U.

b) $\tilde{T}(v + U) = 0$ iff $Tv = 0$ iff $v \in \operatorname{null} T$.

c) $\tilde{T}$ injective iff its null space is $\{0 + U\}$ iff every $v \in \operatorname{null} T$ lies in U iff $\operatorname{null} T \subseteq U$; together with $U \subseteq \operatorname{null} T$, iff $U = \operatorname{null} T$.

d) $\operatorname{null} T = \operatorname{span}\big((1,1,0), e_3\big)$, so $\operatorname{null} \tilde{T} = \operatorname{span}\big((1,1,0) + U\big)$, 1-dimensional. Not injective.

12. The second isomorphism theorem.

a) Linear: S is the quotient map restricted to U. Surjective: $u + w + W = u + W$ because $(u + w) - u \in W$.

b) $u + W = 0 + W$ iff $u \in W$; for $u \in U$ this means $u \in U \cap W$.

c) First isomorphism theorem applied to S. Dimensions: $\dim U - \dim(U \cap W) = \dim(U + W) - \dim W$.

d) $e_2 + e_3 \notin U$, so $U \cap W = \{0\}$ and $U + W = \mathbb{R}^3$ (dimension $2 + 1$). Both sides: $\dim U/\{0\} = 2$ and $\dim \mathbb{R}^3/W = 3 - 1 = 2$.

13. The third isomorphism theorem.

a) $0 + W \in U/W$, and sums and multiples of elements $u + W$ ($u \in U$) are again of this form.
b) Well defined: $v - v' \in W \subseteq U$ gives $v + U = v' + U$. Linear: clear. Surjective: $v + U = S(v + W)$.
c) $S(v + W) = 0 + U$ iff $v \in U$ iff $v + W \in U/W$. The first isomorphism theorem gives $(V/W)/(U/W) \cong V/U$.
d) $\big((x, y, z) + W\big) + U/W \mapsto (x, y, z) + U \mapsto z$, using $V/U \cong \mathbb{R}$ via $(x, y, z) + U \mapsto z$.

14. A quotient operator.

a) $Ae_3 = (0, 0, 3) = 3e_3 \in U$.
b) $Te_1 = (2, 0, 1)$, so $(T/U)(e_1 + U) = 2e_1 + U$. $Te_2 = (1, 2, 1)$, so $(T/U)(e_2 + U) = e_1 + 2e_2 + U$. Matrix $\left(\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right)$.
c) $T|_U = 3I$: eigenvalue 3. T/U: upper-triangular, eigenvalue 2 only; it is not $2I$, so $m_{T/U} = (x - 2)^2$ and T/U is not diagonalizable. T: its eigenvalues are those of $T|_U$ together with those of T/U (Q32): 2 and 3.
d) If T were diagonalizable, T/U would be too (Ex. 60). So T is not diagonalizable.

§3 Duality

15. Dual bases.

a) If $\sum c_j\varphi_j = 0$, evaluate at v_k: $\sum c_j\varphi_j(v_k) = c_k = 0$.
b) n independent vectors in V', which has dimension $\dim \mathcal{L}(V, \mathbb{F}) = n$.
c) Both sides are linear functionals; at v_k the right side is $\sum_j \psi(v_j)\varphi_j(v_k) = \psi(v_k)$. Linear maps that agree on a basis are equal.
d) $B = \left(\begin{smallmatrix} 1 & 3 \\ 2 & 4 \end{smallmatrix}\right)$, $B^{-1} = \left(\begin{smallmatrix} -2 & 3/2 \\ 1 & -1/2 \end{smallmatrix}\right)$: $\varphi_1(x, y) = -2x + \frac{3}{2}y$, $\varphi_2(x, y) = x - \frac{1}{2}y$. Check: $\varphi_1(1, 2) = 1$, $\varphi_1(3, 4) = 0$, $\varphi_2(1, 2) = 0$, $\varphi_2(3, 4) = 1$.

16. Functionals on $\mathcal{P}_2(\mathbb{R})$.

a) For $p = a + b(x - 1) + c(x - 1)^2$: $p(1) = a$, $p'(1) = b$, $\frac{1}{2}p''(1) = c$. So φ_i takes the ith coefficient, which is the defining property of the dual basis.
b) $\psi = \sum \psi(\text{basis vector})\varphi_j$: $\psi(1) = 1$, $\psi(x - 1) = -\frac{1}{2}$, $\psi\big((x - 1)^2\big) = \frac{1}{3}$. So $\psi = \varphi_1 - \frac{1}{2}\varphi_2 + \frac{1}{3}\varphi_3$.
c) $\varepsilon(1) = 1$, $\varepsilon(x - 1) = -1$, $\varepsilon\big((x - 1)^2\big) = 1$: $\varepsilon = \varphi_1 - \varphi_2 + \varphi_3$.
d) Coordinates $(1, 0, 0)$, $(1, -\frac{1}{2}, \frac{1}{3})$, $(1, -1, 1)$. Subtract the first from the others: $(0, -\frac{1}{2}, \frac{1}{3})$, $(0, -1, 1)$; subtract twice the first of these from the second: $(0, 0, \frac{1}{3})$. Rank 3, so they are independent: yes, a basis.

17. The double dual.

a) $j_V(v)(\varphi + \lambda\psi) = (\varphi + \lambda\psi)(v) = \varphi(v) + \lambda\psi(v)$.
b) $j_V(v + \lambda w)(\varphi) = \varphi(v + \lambda w) = \varphi(v) + \lambda\varphi(w)$ for every φ.
c) If $v \neq 0$, extend v to a basis and let φ be 1 at v and 0 at the other basis vectors. Then $j_V(v)(\varphi) = 1$, so $j_V(v) \neq 0$.
d) $\dim V'' = \dim V' = \dim V$, and an injective linear map between spaces of the same finite dimension is an isomorphism.

18. The dual of differentiation.

a) $T1 = 0$, $Tx = 1$, $Tx^2 = 2x$: $\mathcal{M}(T) = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{smallmatrix}\right)$, $\mathcal{M}(T') = \mathcal{M}(T)^T = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 2 \end{smallmatrix}\right)$.
b) For $p = a_0 + a_1x + a_2x^2$, $p' = a_1 + 2a_2x$. $T'(\psi_0)(p) = \psi_0(p') = a_1$, so $T'(\psi_0) = \varphi_1$. $T'(\psi_1)(p) = 2a_2$, so $T'(\psi_1) = 2\varphi_2$.
c) T is surjective, so $\operatorname{null} T' = (\operatorname{range} T)^0 = \{0\}$. $\operatorname{range} T' = \operatorname{span}(\varphi_1, \varphi_2)$, the functionals that are 0 on the constants, which is $(\operatorname{null} T)^0$ since $\operatorname{null} T = \operatorname{span}(1)$.
d) $T'(\varepsilon)(p) = p'(1) = a_1 + 2a_2$: $T'(\varepsilon) = \varphi_1 + 2\varphi_2$.

19. Properties of the dual map.

a) $(S + T)'(\psi) = \psi \circ (S + T) = \psi \circ S + \psi \circ T$; $(cT)'(\psi) = \psi \circ (cT) = c(\psi \circ T)$.
b) $(ST)'(\psi) = \psi \circ S \circ T = (S'\psi) \circ T = T'(S'\psi)$.
c) By (b), $T'(T^{-1})' = (T^{-1}T)' = I_V' = I_{V'}$ and $(T^{-1})'T' = (TT^{-1})' = I_{W'}$.
d) $\mathcal{M}(T) = \left(\begin{smallmatrix} 1 & 1 \\ 2 & 0 \\ 0 & 3 \end{smallmatrix}\right)$, so $\mathcal{M}(T') = \left(\begin{smallmatrix} 1 & 2 & 0 \\ 1 & 0 & 3 \end{smallmatrix}\right)$.

20. Computing annihilators.

a) (a,b,c,d) is 0 on U iff $a+c=0$ and $b+d=0$. Basis: $(1,0,-1,0)$, $(0,1,0,-1)$.

b) $\varepsilon_1, \varepsilon_2 \in U^0$. They are independent: if $a\varepsilon_1 + b\varepsilon_2 = 0$, apply it to $x-2$ to get $-a=0$, and to $x-1$ to get $b=0$. U is the null space of $p \mapsto (p(1), p(2))$, which is surjective, so $\dim U = 2$ and $\dim U^0 = 4-2=2$. Hence $U^0 = \operatorname{span}(\varepsilon_1, \varepsilon_2)$.

c) No: $(\varepsilon_1 - \varepsilon_2)(x) = 1-2 = -1 \neq 0$.

d) $\dim W^0 = 4-2=2$. Example: $p \mapsto p''(0)$ (it is 0 on 1 and x, and 2 on x^2).

21. Null space and range of T'.

a) $T'\psi = 0$ iff $\psi(Tv)=0$ for all v iff ψ is 0 on $\operatorname{range} T$.

b) If $v \in \operatorname{null} T$, $(T'\psi)(v) = \psi(Tv) = \psi(0) = 0$.

c) $\dim \operatorname{range} T' = \dim W - \dim \operatorname{null} T' = \dim W - (\dim W - \operatorname{rank} T) = \operatorname{rank} T = \dim V - \dim \operatorname{null} T = \dim (\operatorname{null} T)^0$. A subspace of the same dimension is the whole space.

d) T surjective iff $\operatorname{range} T = W$ iff $(\operatorname{range} T)^0 = \{0\}$ iff $\operatorname{null} T' = \{0\}$. From (c), $\operatorname{rank} T' = \operatorname{rank} T$; with $\mathcal{M}(T') = \mathcal{M}(T)^T$, $\operatorname{rank} A^T = \operatorname{rank} A$.

22. Annihilators in general.

a) Surjective: a functional on U extends to V (extend a basis of U, set it to 0 on the new vectors). Null space: $\varphi|_U = 0$ iff $\varphi \in U^0$. Rank–nullity: $\dim V' = \dim U^0 + \dim U'$, so $\dim U^0 = \dim V - \dim U$.

b) If $U \subseteq W$, anything 0 on W is 0 on U. Conversely, if $u \in U \setminus W$, then a basis of W together with u is independent; extend it to a basis of V and let φ be 1 at u and 0 on the other basis vectors. Then $\varphi \in W^0$ but $\varphi \notin U^0$.

c) φ is 0 on all $u+w$ iff it is 0 on U (take $w=0$) and on W (take $u=0$).

d) $v \mapsto \big(\varphi_1(v), \dots, \varphi_m(v)\big)$ is a linear map $V \to \mathbb{F}^m$ with null space $\bigcap \operatorname{null} \varphi_i = \{0\}$, so it is injective and $\dim V \leq m$.

23. When functionals form a basis.

a) Let $\eta_1, \dots, \eta_n$ be the coordinate functionals on $\mathbb{F}^n$. $S'(\eta_i) = \eta_i \circ S = \varphi_i$, so $S'\big(\sum c_i \eta_i\big) = \sum c_i \varphi_i$. Hence S' is injective iff the φ_i are independent. And S' is injective iff S is surjective (Prop. 51).

b) n functionals in the n-dimensional V' form a basis iff they are independent iff S is surjective iff S is injective ($\dim V = n = \dim \mathbb{F}^n$) iff $\operatorname{null} S = \{0\}$.

c) A polynomial of degree at most 2 with roots $0, 1, 2$ is 0. By (b), yes.

§4 Invariant subspaces and eigenvalues

24. Invariant subspaces of a 3×3 matrix.

a) The matrix $\left(\begin{smallmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 5 \end{smallmatrix}\right)$ is lower-triangular, so in the reversed basis e_3, e_2, e_1 it is upper-triangular with the same diagonal (Prop. 60): eigenvalues $2, 5$. $E(2,T)$: $(T-2I)(x,y,z) = (0,x,3z) = 0$ iff $x=z=0$: $\operatorname{span}(e_2)$. $E(5,T) = \operatorname{span}(e_3)$.

b) $Te_1 = (2,1,0) \notin \operatorname{span}(e_1)$: not invariant. $\operatorname{span}(e_1, e_2)$: $Te_1 = 2e_1 + e_2$, $Te_2 = 2e_2$: invariant. $\operatorname{span}(e_2, e_3)$: $Te_2 = 2e_2$, $Te_3 = 5e_3$: invariant.

c) An invariant line is spanned by an eigenvector: $\operatorname{span}(e_2)$ and $\operatorname{span}(e_3)$ only.

d) $\dim E(2,T) + \dim E(5,T) = 2 < 3$: not diagonalizable.

25. Existence of eigenvalues.

a) $v, Tv, \dots, T^n v$ are $n+1$ vectors in an n-dimensional space, so $\sum_{k=0}^{n} a_k T^k v = 0$ with not all $a_k = 0$; take $p = \sum a_k x^k$. It is not a nonzero constant, since $a_0 v = 0$ with $v \neq 0$ forces $a_0 = 0$.

b) Over $\mathbb{C}$ the nonconstant p has a root λ (Fundamental Theorem of Algebra), so $p = (x-\lambda)q$ with $\deg q = \deg p - 1$. If $q(T)v = 0$, then q (nonzero, since $p \neq 0$) would be a smaller-degree polynomial with this property, contradicting minimality.

c) $(T-\lambda I)w = (T-\lambda I)q(T)v = p(T)v = 0$ and $w \neq 0$.

d) For R, $v = e_1$: $R^2 e_1 = -e_1$, so the minimal p is x^2+1, which has no real root; step (b) fails.

26. Independence of eigenvectors.

a) Suppose not; let k be smallest with $v_k = \sum_{i<k} a_i v_i$. Apply $T - \lambda_k I$: $0 = \sum_{i<k} a_i(\lambda_i - \lambda_k)v_i$. The v_i, $i < k$, are independent and $\lambda_i \neq \lambda_k$, so all $a_i = 0$, so $v_k = 0$, impossible for an eigenvector.
b) One eigenvector per eigenvalue gives an independent list, of length at most $\dim V$.
c) They are eigenvectors of differentiation (on infinitely differentiable functions) with distinct eigenvalues $1, 2, 3$.
d) $(1, \lambda, \lambda^2, \dots) \mapsto (\lambda, \lambda^2, \dots) = \lambda(1, \lambda, \dots)$. No contradiction: the space of sequences is infinite-dimensional.

27. A reflection of polynomials.

a) $T^2p(x) = p(-(-x)) = p(x)$. If $Tp = \lambda p$, $p \neq 0$, then $p = T^2p = \lambda^2 p$, so $\lambda^2 = 1$.
b) $Tp = p$ iff p is even: $E(1, T) = \operatorname{span}(1, x^2)$. $Tp = -p$ iff p is odd: $E(-1, T) = \operatorname{span}(x, x^3)$.
c) Dimensions $2 + 2 = 4 = \dim V$. Or: m_T divides $x^2 - 1 = (x - 1)(x + 1)$, which has distinct roots.
d) Same eigenvalues ± 1, each with eigenspace of dimension 2 (eigenvalues of the dual map).

28. Diagonalizing a 2×2 matrix.

a) $\chi_A(t) = t^2 - 5t + 6 = (t - 2)(t - 3)$. $A - 2I = \left(\begin{smallmatrix} 2 & -2 \\ 1 & -1 \end{smallmatrix}\right)$: eigenvector $(1, 1)$. $A - 3I = \left(\begin{smallmatrix} 1 & -2 \\ 1 & -2 \end{smallmatrix}\right)$: eigenvector $(2, 1)$.
b) $P = \left(\begin{smallmatrix} 1 & 2 \\ 1 & 1 \end{smallmatrix}\right)$, $D = \operatorname{diag}(2, 3)$, $P^{-1} = \left(\begin{smallmatrix} -1 & 2 \\ 1 & -1 \end{smallmatrix}\right)$.
c) $A^k = PD^kP^{-1} = \left(\begin{smallmatrix} -2^k + 2 \cdot 3^k & 2^{k+1} - 2 \cdot 3^k \\ -2^k + 3^k & 2^{k+1} - 3^k \end{smallmatrix}\right)$. Check $k = 1$: A.
d) Distinct eigenvalues $2, 3$, diagonalizable: $m_A = (x - 2)(x - 3)$.

29. Eigenvalues that survive changes.

a) $Rv = \lambda v \iff S^{-1}TSv = \lambda v \iff T(Sv) = \lambda(Sv)$, and $Sv \neq 0 \iff v \neq 0$.
b) $(T - \lambda I)' = T' - \lambda I$, and a map and its dual have the same rank; V and V' have the same dimension, so the null spaces have the same dimension.
c) $E(1, A) = \operatorname{span}(e_1)$ and $E(1, A^T) = \operatorname{span}(e_2)$: same dimension, different subspaces.

30. Stochastic matrices.

a) With $\|x\|_1 = \sum |x_i|$: $\|Px\|_1 \leq \sum_i \sum_j P_{ij}|x_j| = \sum_j |x_j|$ (columns sum to 1). If $Px = \lambda x$, $x \neq 0$: $|\lambda|\|x\|_1 \leq \|x\|_1$.
b) Column sums 1 mean $\mathbf{1}^T P = \mathbf{1}^T$, i.e. $P^T\mathbf{1} = \mathbf{1}$. So 1 is an eigenvalue of P^T, and P and P^T have the same eigenvalues (the matrix of the dual map is the transpose).
c) $(P - I)\pi = 0$: $-\frac{1}{2}\pi_1 + \frac{1}{5}\pi_2 = 0$, so $\pi_2 = \frac{5}{2}\pi_1$. Normalizing: $\pi = (\frac{2}{7}, \frac{5}{7})$.
d) $\chi_P(t) = t^2 - \frac{13}{10}t + \frac{3}{10} = (t - 1)(t - \frac{3}{10})$: the other eigenvalue is $\frac{3}{10}$.

31. PageRank.

a) If $(I - \alpha P)x = 0$ with $x \neq 0$, then $Px = \frac{1}{\alpha}x$ with $\frac{1}{\alpha} > 1$, contradicting Q30(a). So $I - \alpha P$ is injective, hence invertible.
b) $r_1 = \frac{1}{2}r_2 + \frac{1}{2}$ and $r_2 = \frac{1}{2}r_1$. So $r_1 = \frac{1}{4}r_1 + \frac{1}{2}$, $r_1 = \frac{2}{3}$, $r_2 = \frac{1}{3}$.
c) Entries are nonnegative and add to 1. All teleports go to page 1, and page 1 also receives links from page 2, so page 1 ranks higher.

32. Eigenvalues of restriction and quotient.

a) An eigenvector of $T|_U$ is a nonzero $u \in U$ with $Tu = \lambda u$.
b) $v \notin U$ means $v + U \neq 0 + U$, and $(T/U)(v + U) = Tv + U = \lambda(v + U)$.
c) $T - \lambda I$ is injective, so its restriction to U is injective; it maps U into U (invariance), so onto U (same finite dimension). If $\big((T/U) - \lambda I\big)(v + U) = 0 + U$, then $(T - \lambda I)v \in U$, so $(T - \lambda I)v = (T - \lambda I)u$ for some $u \in U$, and injectivity gives $v = u \in U$, i.e. $v + U = 0 + U$. So λ is not an eigenvalue of T/U.

§5 Upper-triangular matrices and diagonalizability

33. Upper-triangular matrices.

a) Column j of $\mathcal{M}(T)$ holds the coefficients of Tv_j; zeros below row j mean $Tv_j = \sum_{i \leq j} a_{ij} v_i$.
b) For $i \leq j$, $Tv_i \in \operatorname{span}(v_1, \dots, v_i) \subseteq \operatorname{span}(v_1, \dots, v_j)$, so T maps a spanning list of $\operatorname{span}(v_1, \dots, v_j)$ into it.
c) $v_j \in \operatorname{span}(v_1, \dots, v_j)$, which is invariant, so $Tv_j \in \operatorname{span}(v_1, \dots, v_j)$: the entries of column j below row j are 0.

34. Eigenvalues on the diagonal.

a) $\mathcal{M}(T-\lambda I)$ is upper-triangular with diagonal $\lambda_k - \lambda$. If $\lambda \neq$ every λ_k, back-substitution in $(T-\lambda I)x = 0$ (last row first) gives $x = 0$: injective, not an eigenvalue. If $\lambda = \lambda_k$, the first k columns of $\mathcal{M}(T-\lambda I)$ have zeros in rows $k, \dots, n$, so they are k vectors in a $(k-1)$-dimensional space, hence dependent: not injective.

b) T invertible iff 0 is not an eigenvalue iff no $\lambda_k = 0$.

c) In $1, x, x^2$: $\mathcal{M}(T) = \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, so the only eigenvalue is 1. $E(1,T) = \operatorname{null} D = \operatorname{span}(1)$.

d) $D^3 = 0$ on $\mathcal{P}_2(\mathbb{R})$, and $(I+D)(I-D+D^2) = I + D^3 = I$. So $T^{-1} = I - D + D^2$.

35. Triangularizing a matrix.

a) $\chi_A(t) = t^2 - 2t + 1 = (t-1)^2$: eigenvalue 1. $A - I = \left(\begin{smallmatrix} 2 & 1 \\ -4 & -2 \end{smallmatrix}\right)$: eigenvectors are the multiples of $(1, -2)$.

b) $s_1 = (1, -2)$, $s_2 = e_2$. $As_1 = s_1$ and $As_2 = (1, -1) = s_1 + s_2$. So with $S = \left(\begin{smallmatrix} 1 & 0 \\ -2 & 1 \end{smallmatrix}\right)$, $S^{-1}AS = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$.

c) $\dim E(1, A) = 1 < 2$, so not diagonalizable. $(A-I)^2 = 0$ (multiply out) and $A \neq I$, so $m_A = (x-1)^2$.

36. A diagonalizability test with a parameter.

a) Lower-triangular with diagonal $1, 2, 2$; in the reversed basis e_3, e_2, e_1 it is upper-triangular (Prop. 60): eigenvalues 1 and 2.

b) $E(1)$ has dimension 1 ($A_t - I$ has rank 2). $A_t - 2I = \left(\begin{smallmatrix} -1 & 0 & 0 \\ 1 & 0 & 0 \\ -3 & t & 0 \end{smallmatrix}\right)$ has rank 1 if $t = 0$ and rank 2 if $t \neq 0$. So $\dim E(1) + \dim E(2) = 3$ iff $t = 0$: diagonalizable exactly for $t = 0$.

c) $t = 0$: $E(1)$ is the null space of $\left(\begin{smallmatrix} 0 & 0 & 0 \\ 1 & 1 & 0 \\ -3 & 0 & 1 \end{smallmatrix}\right)$, spanned by $(1, -1, 3)$. $E(2) = \{x = 0\} = \operatorname{span}(e_2, e_3)$. Eigenbasis $(1, -1, 3),\ e_2,\ e_3$.

37. Conditions for diagonalizability.

a) Column j of a diagonal matrix has only the (j, j) entry, i.e. $Tv_j = a_{jj}v_j$: each basis vector is an eigenvector, and conversely.

b) If $w_1 + \cdots + w_m = 0$ with $w_i \in E(\lambda_i, T)$, the nonzero w_i would be eigenvectors for distinct eigenvalues with a nontrivial dependence, contradicting independence (Prop. 54). So all $w_i = 0$.

c) ($\Rightarrow$) The eigenbasis lies in the sum, so the (direct) sum is V. ($\Leftarrow$) Put together bases of the $E(\lambda_i, T)$.

d) The direct sum has dimension $\sum \dim E(\lambda_i, T)$, and a subspace equals V iff its dimension is $\dim V$.

38. Projections.

a) $v = (v - Pv) + Pv$ with $P(v - Pv) = Pv - P^2v = 0$. If $v = Pw \in \operatorname{range} P$ and $Pv = 0$, then $v = Pw = P^2w = Pv = 0$.

b) If $v = Pw$ then $Pv = P^2w = Pw = v$; if $Pv = v$ then $v \in \operatorname{range} P$.

c) $V = E(0, P) \oplus E(1, P)$ by (a),(b), so P is diagonalizable. m_P divides $x(x-1)$: it is x ($P = 0$), $x - 1$ ($P = I$), or $x(x-1)$ (otherwise).

d) $(x, y) = y(1, 1) + (x - y)(1, 0)$, so $P(x, y) = (y, y)$: matrix $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}\right)$.

39. A recurrence.

a) $A = \left(\begin{smallmatrix} 0 & 1 \\ 2 & 1 \end{smallmatrix}\right)$, $\chi_A(t) = t^2 - t - 2 = (t-2)(t+1)$. Eigenvectors: $(1, 2)$ for 2, $(1, -1)$ for -1.

b) $(0, 1) = \frac{1}{3}(1, 2) - \frac{1}{3}(1, -1)$.

c) $A^n(0, 1) = \frac{1}{3}\big(2^n(1, 2) - (-1)^n(1, -1)\big)$; the first entry is $a_n = \dfrac{2^n - (-1)^n}{3}$. Check: $a_2 = \frac{4-1}{3} = 1$, $a_3 = \frac{8+1}{3} = 3$; and indeed $a_2 = 1 + 0 = 1$, $a_3 = 1 + 2 = 3$.

40. A two-state Markov chain.

a) $\chi_P(t) = t^2 - 1.7t + 0.7 = (t-1)(t-0.7)$. Eigenvectors: $(2, 1)$ for 1 ($-0.1x + 0.2y = 0$), $(1, -1)$ for 0.7.

b) $\pi = (\frac{2}{3}, \frac{1}{3})$, and $(1, 0) = \pi + \frac{1}{3}(1, -1)$.

c) $P^k p_0 = \pi + \frac{1}{3}(0.7)^k(1, -1)$.

d) The limit is $\pi = (\frac{2}{3}, \frac{1}{3})$; the error is multiplied by 0.7 at each step.

§6 The minimal polynomial

41. Minimal polynomials from the definition.

a) A is not a multiple of I, so I, A are independent. $A^2 = \left(\begin{smallmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right) = 2A - I$. So $m_A = x^2 - 2x + 1 = (x-1)^2$.

b) B is not a multiple of I; $(B - I)(B - 2I) = \operatorname{diag}(0, 0, 0)$, so $m_B = (x-1)(x-2)$.

c) B (distinct roots); not A (repeated root).

d) $m_{A-I}(x) = m_A(x+1) = x^2$. m_{2B}: $2B = \operatorname{diag}(2, 2, 4)$, so $m_{2B} = (x-2)(x-4)$.

42. Polynomials that annihilate T.

a) If $q = m_T r$, then $q(T) = m_T(T)r(T) = 0$. Conversely, divide: $q = m_T r + s$ with $s = 0$ or $\deg s < \deg m_T$. Then $s(T) = q(T) - m_T(T)r(T) = 0$, and minimality forces $s = 0$.

b) $x^4 - x = x(x-1)(x^2 + x + 1)$ has the distinct roots $0, 1, \omega, \bar{\omega}$ ($\omega = e^{2\pi i/3}$). m_T divides it, so m_T has distinct roots and T is diagonalizable; eigenvalues lie among $0, 1, \omega, \bar{\omega}$.

c) Rotation by 120°: $R^3 = I$, so $R^4 = R$, but R has no real eigenvalue, so it is not diagonalizable over $\mathbb{R}$. Real eigenvalues must be real roots of $x^4 - x$: only 0 and 1.

43. The degree bound.

a) $v, \ldots, T^n v$ are dependent, giving nonzero q of degree $\leq n$ with $q(T)v = 0$. If $q(T) = 0$, then m_T divides q and $\deg m_T \leq n$.

b) If $q(T)u = 0$ then $q(T)Tu = Tq(T)u = 0$. $v \in U$, so $U \neq \{0\}$; $q(T) \neq 0$, so $U \neq V$.

c) $m_{T/U}(T)v \in U$ for every v (since $m_{T/U}(T/U) = 0$ and $p(T/U)(v + U) = p(T)v + U$), and $m_{T|_U}(T)$ sends U to 0. So $m_{T|_U}(T)m_{T/U}(T) = 0$, and Q42(a) applies.

d) Base case $n = 1$: $V = \operatorname{span}(v)$, so $q(T)v = 0$ gives $q(T) = 0$ and (a) applies. Inductive step: by induction $\deg m_{T|_U} \leq \dim U$ and $\deg m_{T/U} \leq \dim V/U$. So $\deg m_T \leq \dim U + \dim V/U = n$.

44. Distinct roots give diagonalizability.

a) Divide g by $x - \lambda_1$: the remainder s has degree < 1, so it is a constant, and $s = g(\lambda_1) = \prod_{i \geq 2}(\lambda_1 - \lambda_i) \neq 0$.

b) $(T - \lambda_1 I)u = \frac{1}{s} m_T(T)v = 0$. And $v - u = \frac{1}{s}\big(sv - g(T)v\big) = -\frac{1}{s}(T - \lambda_1 I)q(T)v \in \operatorname{range}(T - \lambda_1 I)$.

c) By (b) V is the sum. By rank–nullity the dimensions add to $\dim V$, so the sum is direct.

d) $U = \operatorname{range}(T - \lambda_1 I)$ is invariant, $m_{T|_U}$ divides m_T (so has distinct roots), and $\dim U < \dim V$ since λ_1 is an eigenvalue. By induction $T|_U$ has an eigenbasis; add a basis of $\operatorname{null}(T - \lambda_1 I)$. (Base case: $m = 1$ means $T = \lambda_1 I$.)

45. A companion matrix.

a) $Ae_1 = (0, 0, 6)$ (the first column), $A^2 e_1 = A(0, 0, 6) = (0, 6, 36)$, $A^3 e_1 = A(0, 6, 36) = (6, 36, 150)$. The vectors $e_1, (0, 0, 6), (0, 6, 36)$ are independent (look at the positions of the first nonzero entries), so they form a basis. If $q(A) = 0$ with $\deg q \leq 2$, then $q(A)e_1 = 0$ is a dependence among $e_1, Ae_1, A^2 e_1$, so $q = 0$; hence $\deg m_A \geq 3$. Solving $A^3 e_1 = ae_1 + bAe_1 + cA^2 e_1 = (a, 6c, 6b + 36c)$ gives $a = 6$, $c = 6$, $b = -11$. So $p(x) = x^3 - 6x^2 + 11x - 6$ has $p(A)e_1 = 0$, hence $p(A)A^j e_1 = A^j p(A)e_1 = 0$ for each basis vector, so $p(A) = 0$ and $m_A = p = (x-1)(x-2)(x-3)$.

b) Yes: m_A splits over $\mathbb{R}$ with distinct roots.

c) $A(A^2 - 6A + 11I) = 6I$, so $A^{-1} = \frac{1}{6}(A^2 - 6A + 11I)$.

d) $x^4 = (x + 6)m_A(x) + 25x^2 - 60x + 36$, so $A^4 = 25A^2 - 60A + 36I$.

46. Minimal polynomial of a direct sum.

a) Write $v = u + w$. Then $p(T)v = p(T|_U)u + p(T|_W)w$ with the two terms in U and W. So $p(T) = 0$ iff both restrictions give 0 (the sum is direct).

b) By (a) and Q42(a), $p(T) = 0$ iff $m_{T|_U}$ and $m_{T|_W}$ both divide p. m_T is the monic polynomial of least degree with $m_T(T) = 0$, so it is the monic polynomial of least degree divisible by both.

c) If both restrictions have minimal polynomials with distinct linear factors, so does their least common multiple, and conversely each divides m_T.

d) The blocks have minimal polynomials $(x-1)^2$ and $(x-1)(x-2)$. Least common multiple: $m_T = (x-1)^2(x-2)$.

47. An operator on polynomials.

a) $T(x^k) = kx^k$: $\mathcal{M}(T) = \operatorname{diag}(0, 1, 2, 3)$. Eigenvalues $0, 1, 2, 3$.

b) Diagonalizable with distinct eigenvalues: $m_T = x(x-1)(x-2)(x-3)$.

c) $T^2 = \operatorname{diag}(0, 1, 4, 9)$: $m_{T^2} = x(x-1)(x-4)(x-9)$.

d) The Lagrange polynomial that is 1 at 2 and 0 at $0, 1, 3$: $p(x) = \dfrac{x(x-1)(x-3)}{2 \cdot 1 \cdot (-1)} = -\frac{1}{2}x(x-1)(x-3)$. Then $p(T)x^k = p(k)x^k$, which is x^2 for $k = 2$ and 0 otherwise.

48. Real operators.

a) Factor m_T over $\mathbb{R}$. If it has a factor $x - \lambda$, then λ is an eigenvalue and an eigenvector spans an invariant line. Otherwise $m_T = (x^2 + bx + c)q$; minimality gives v with $w = q(T)v \neq 0$, and $(T^2 + bT + cI)w = 0$, so $\operatorname{span}(w, Tw)$ is invariant, of dimension 1 or 2.

b) Dimension 1: clear. If $\dim V \geq 3$ is odd and T has no eigenvalue, (a) gives an invariant U of dimension 2. Then $\dim V/U$ is odd and smaller, so T/U has an eigenvalue λ, a root of $m_{T/U}$, which divides m_T; so λ is a root of m_T and an eigenvalue of T. Contradiction.

c) $T^2 = -I$, so an eigenvalue would satisfy $\lambda^2 = -1$: none real. $\operatorname{span}(e_1, e_2)$ is invariant ($Te_1 = e_2$, $Te_2 = -e_1$).

§7 Quick true or false

1. True: $(v, w) \mapsto (w, v)$.
2. False: three different lines in $\mathbb{R}^2$.
3. True: $0 \in v + U$ iff $-v \in U$ iff $v \in U$.
4. True: $\pi\colon V \to V/\{0\}$ is injective; and every $v + V = 0 + V$.
5. True: first isomorphism theorem with range $T = W$.
6. False: $U = \operatorname{span}(e_1)$, $W = \operatorname{span}(e_2)$ in $\mathbb{R}^2$; both quotients are 1-dimensional.
7. True: second isomorphism theorem.
8. True: given $S\colon V/U \to W$, take $T = S \circ \pi$.
9. True: φ is surjective onto $\mathbb{F}$; rank–nullity.
10. False: T injective iff T' surjective. The inclusion $\mathbb{F} \to \mathbb{F}^2$ is injective and its dual is not.
11. False: it is $U^0 + W^0$. For two different lines U, W in $\mathbb{R}^2$, $(U \cap W)^0 = V'$ but $U^0 \cap W^0 = \{0\}$.
12. False: in $\mathbb{R}^2$ with $v_1 = e_1$, $v_2 = e_2$, $\varphi_1 = x$; with $v_1 = e_1$, $v_2 = (1, 1)$, $\varphi_1 = x - y$.
13. False: rotation by 90°.
14. False: $T = I$ has no eigenvalue 0.
15. False: e_1 and e_2 for $\operatorname{diag}(1, 2)$; their sum is not an eigenvector.
16. True: $T^2 v = \lambda^2 v$.
17. False: I.
18. True: in an eigenbasis T^{-1} is diagonal with entries $1/\lambda_i$.
19. True: $m_{T|_U}$ divides m_T.
20. False: rotation by 90° on $\mathbb{R}^2$ (no eigenvalue, so no first basis vector can be an eigenvector).
21. True: Prop. 67.
22. False: $T = I$, $p = (x-1)(x-2)$.
23. True: $m_T(0) = 0$ would make 0 an eigenvalue.
24. False: $\operatorname{diag}(1, 1, 2)$.